Amazon SageMaker

Bring Your Own Model

Immersion Day

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Model Options

Training code

AWS Marketplace for Machine Learning

Amazon SageMaker AutoPilot

- XGBoost - Gradient Boosted Trees
- Matrix Factorization
- Regression
- Principal Component Analysis
- K-Means Clustering
- And More!

Built-in Algorithms (17)
No ML coding required

TensorFlow, Keras, mxnet, PYTORCH, HOROVOD, GLUON, dmlc XGBoost, APACHE Spark, Chainer, scikit learn

Bring Your Own Script
Amazon SageMaker builds the ccontainer
Open source containers

Bring Your Own Container
Full control, you build the container
R, C++, etc

Fully Managed, Distributed, Auto-Scaled, Secured

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aws machine learning

Containers for Machine Learning

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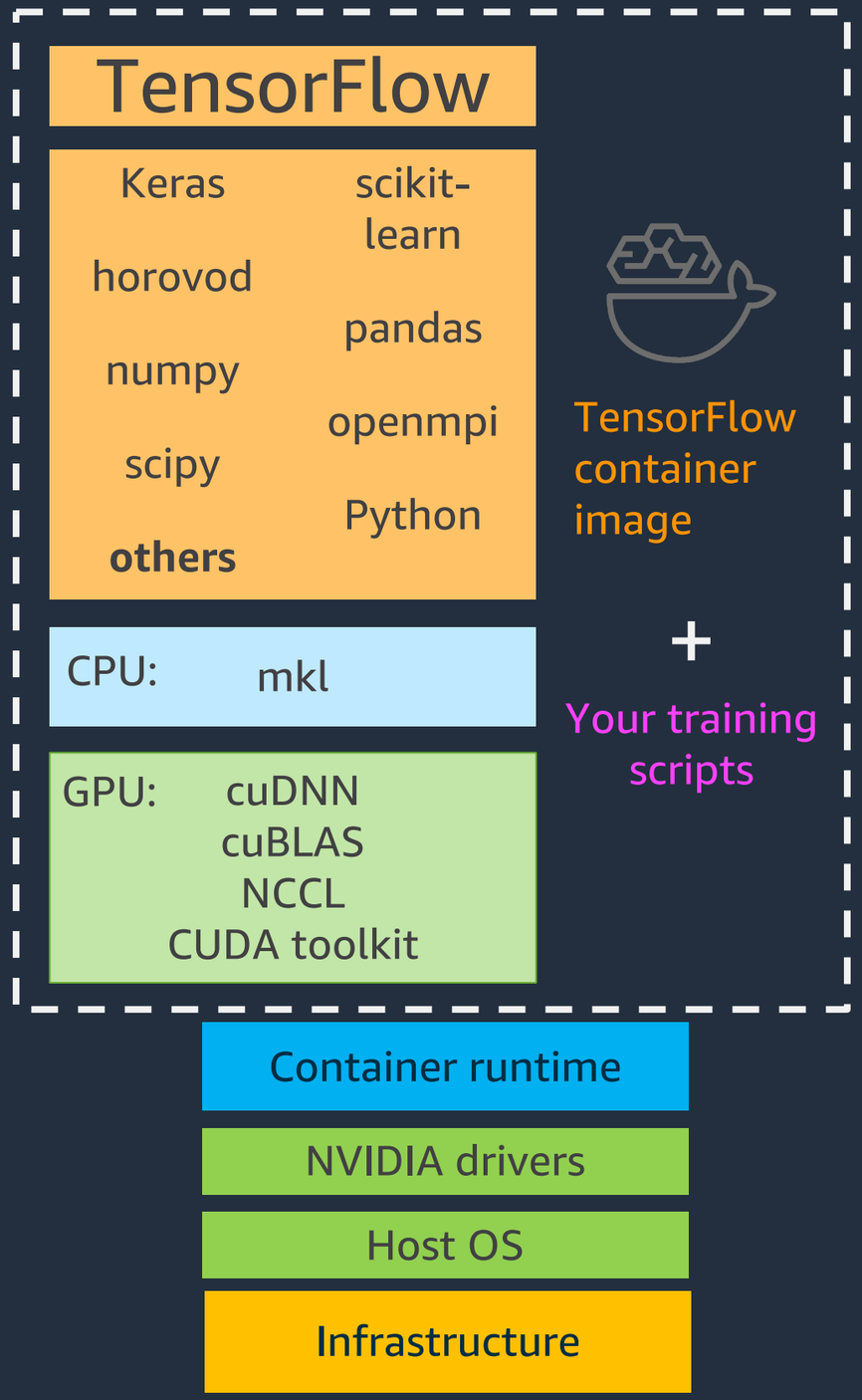
Containers for machine learning



ML environments that are:

- Lightweight
- Portable
- Scalable
- Consistent

Packages:

- Training code
- Dependencies
- Configurations

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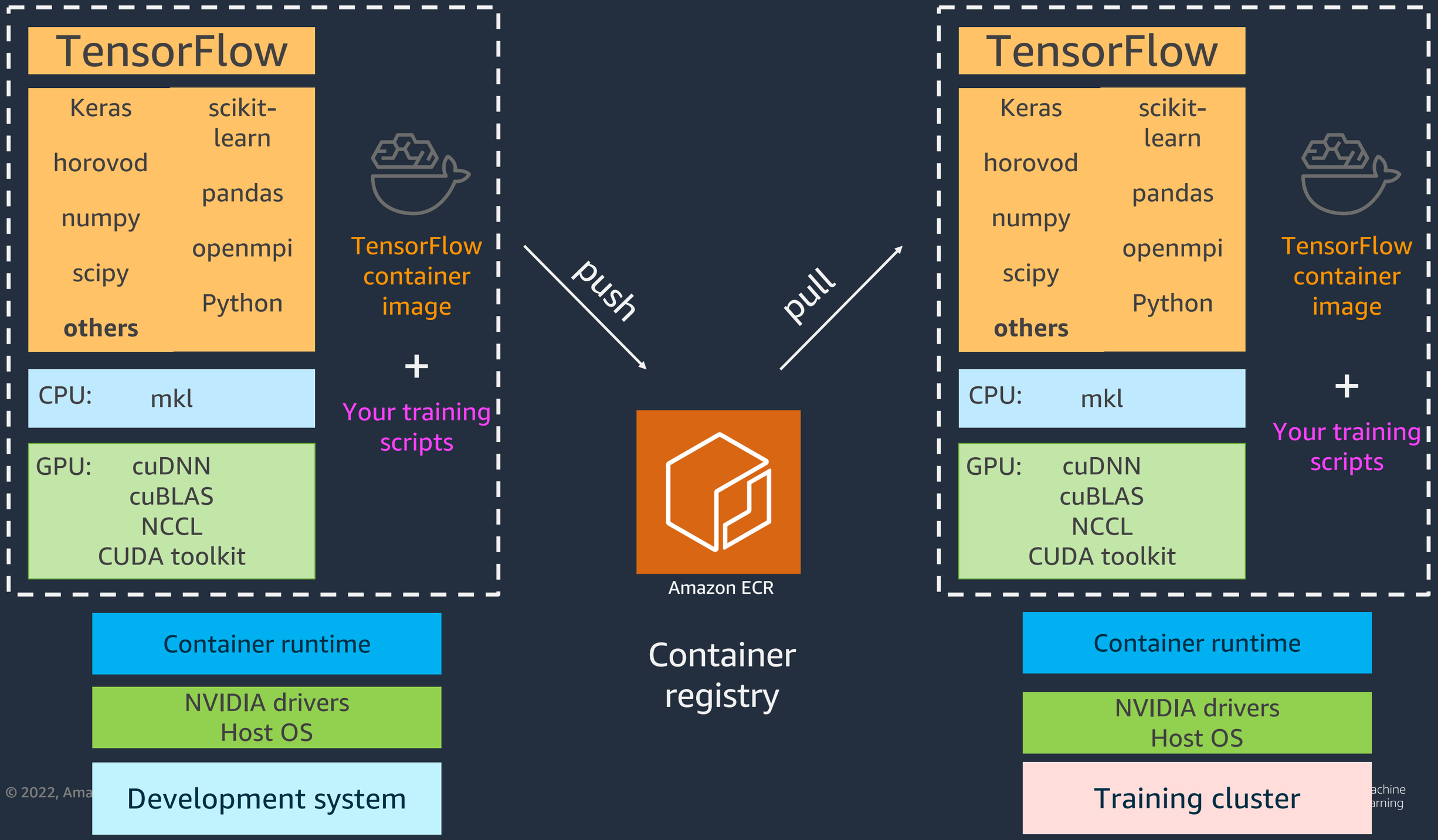
TensorFlow
Keras
horovod
numpy
scipy
others
scikit-learn
pandas
openmpi
Python
CPU: mkl
GPU: cuDNN
cuBLAS
NCCL
CUDA toolkit
TensorFlow container image
+
Your training scripts
push
pull
Amazon ECR
Container registry
TensorFlow
Keras
horovod
numpy
scipy
others
scikit-learn
pandas
openmpi
Python
CPU: mkl
GPU: cuDNN
cuBLAS
NCCL
CUDA toolkit
TensorFlow container image
+
Your training scripts
Container runtime
NVIDIA drivers
Host OS
Development system
Container runtime
NVIDIA drivers
Host OS
Training cluster

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AWS Deep Learning Containers

- Prepackaged Docker container images fully configured and validated
- Optimized for performance with latest NVIDIA driver, CUDA libraries, and Intel libraries
- Consistent and reproducible deployment and lightweight
- Optimized for distributed machine learning
- Runs on Amazon ECS, Amazon EKS and Amazon SageMaker

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AWS Deep Learning Container Example (PyTorch 1.6)

The SageMaker Training Toolkit is an open-source library already installed in AWS Deep Learning Containers.

```
# The following section uninstalls torch and torchvision before installing the
# custom versions from an S3 bucket. This will need to be removed in the future
RUN pip install --upgrade pip --trusted-host pypi.org --trusted-host files.pythonhosted.org \
 && ln -s /opt/conda/bin/pip /usr/local/bin/pip3 \
 && pip install --no-cache-dir -U \
    "awscli<2" \
    fastai==1.0.59 \
    scipy==1.2.2 \
    click \
    "cryptography>3.2" \
    smdebug==${SMDEBUG_VERSION} \
    "sagemaker>=2,<3" \
    sagemaker-experiments==0.* \
    "sagemaker-pytorch-training<3" \
    psutil==5.6.7 \
    Pillow==7.1.0 \
 && pip install --no-cache-dir -U ${PT_TRAINING_URL} \
 && pip uninstall -y torchvision \
 && pip install --no-deps --no-cache-dir -U \
    https://torchvision-build.s3.amazonaws.com/1.6.0/cpu/torchvision-0.7.0-cp36-cp36m-linux_x86_64.whl
```

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Bring your own Script

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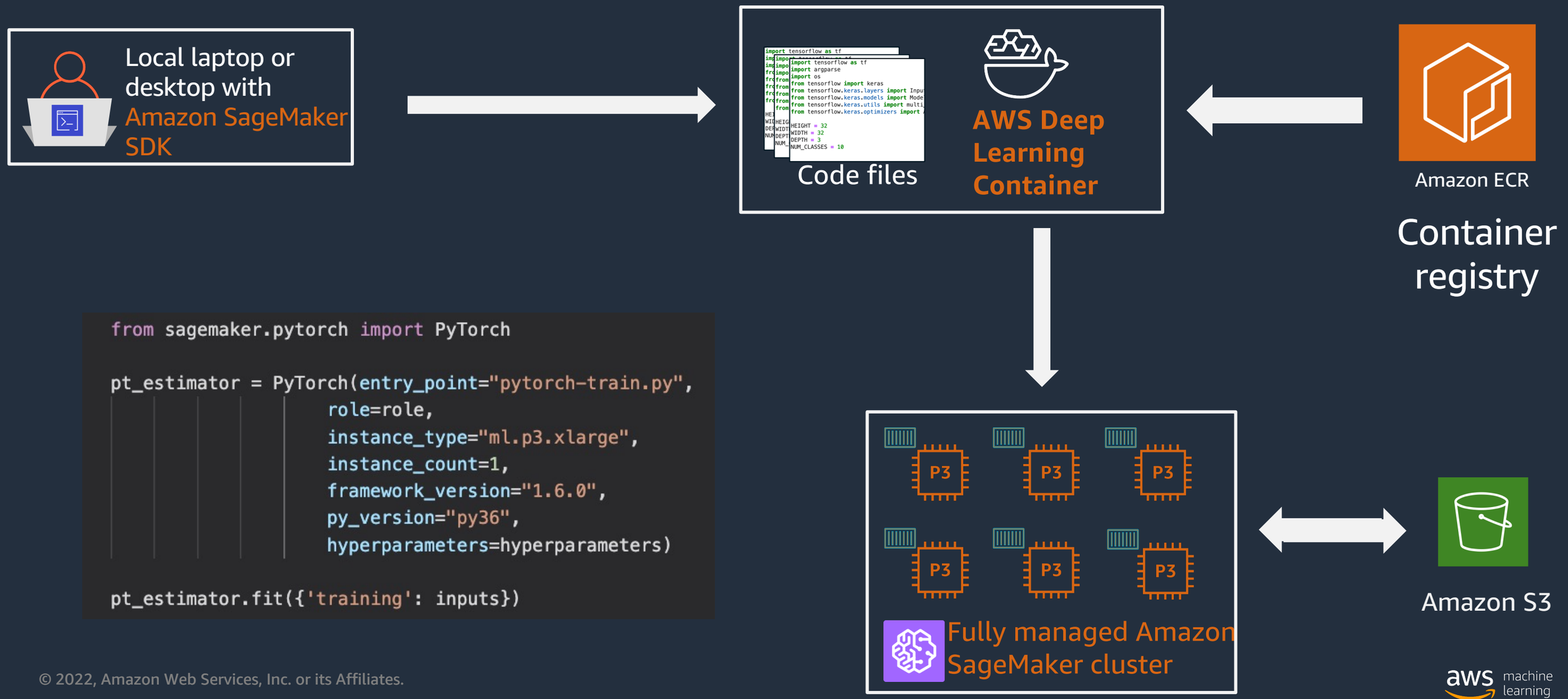
High Level Workflow




```
from sagemaker.pytorch import PyTorch

pt_estimator = PyTorch(entry_point="pytorch-train.py",
                       role=role,
                       instance_type="ml.p3.xlarge",
                       instance_count=1,
                       framework_version="1.6.0",
                       py_version="py36",
                       hyperparameters=hyperparameters)

pt_estimator.fit({'training': inputs})
```

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Bring Your Own Script

AWS
Managed

1. Point to the AWS-managed container of your choice
2. Write your model as a bundle of files.
3. Specify the entry-point in the SageMaker Estimator
4. Include any extra libraries with requirements.txt
5. Use our web-server for inference

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Changes required to your script

```python
import argparse
import os

if __name__ =='__main__':

    parser = argparse.ArgumentParser()

    # hyperparameters sent by the client are passed as command-line arguments to the script.
    parser.add_argument('--epochs', type=int, default=50)
    parser.add_argument('--batch-size', type=int, default=64)
    parser.add_argument('--learning-rate', type=float, default=0.05)
    parser.add_argument('--use-cuda', type=bool, default=False)

    # Data, model, and output directories
    parser.add_argument('--output-data-dir', type=str, default=os.environ['SM_OUTPUT_DATA_DIR'])
    parser.add_argument('--model-dir', type=str, default=os.environ['SM_MODEL_DIR'])
    parser.add_argument('--train', type=str, default=os.environ['SM_CHANNEL_TRAIN'])
    parser.add_argument('--test', type=str, default=os.environ['SM_CHANNEL_TEST'])

    args, _ = parser.parse_known_args()

    # ... load from args.train and args.test, train a model, write model to args.model_dir.
```

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Upload data to S3

- Training data uploaded to: s3://my-data-bucket/path/to/my/training/data
- Validation data uploaded to: s3://my-data-bucket/path/to/my/validation/data

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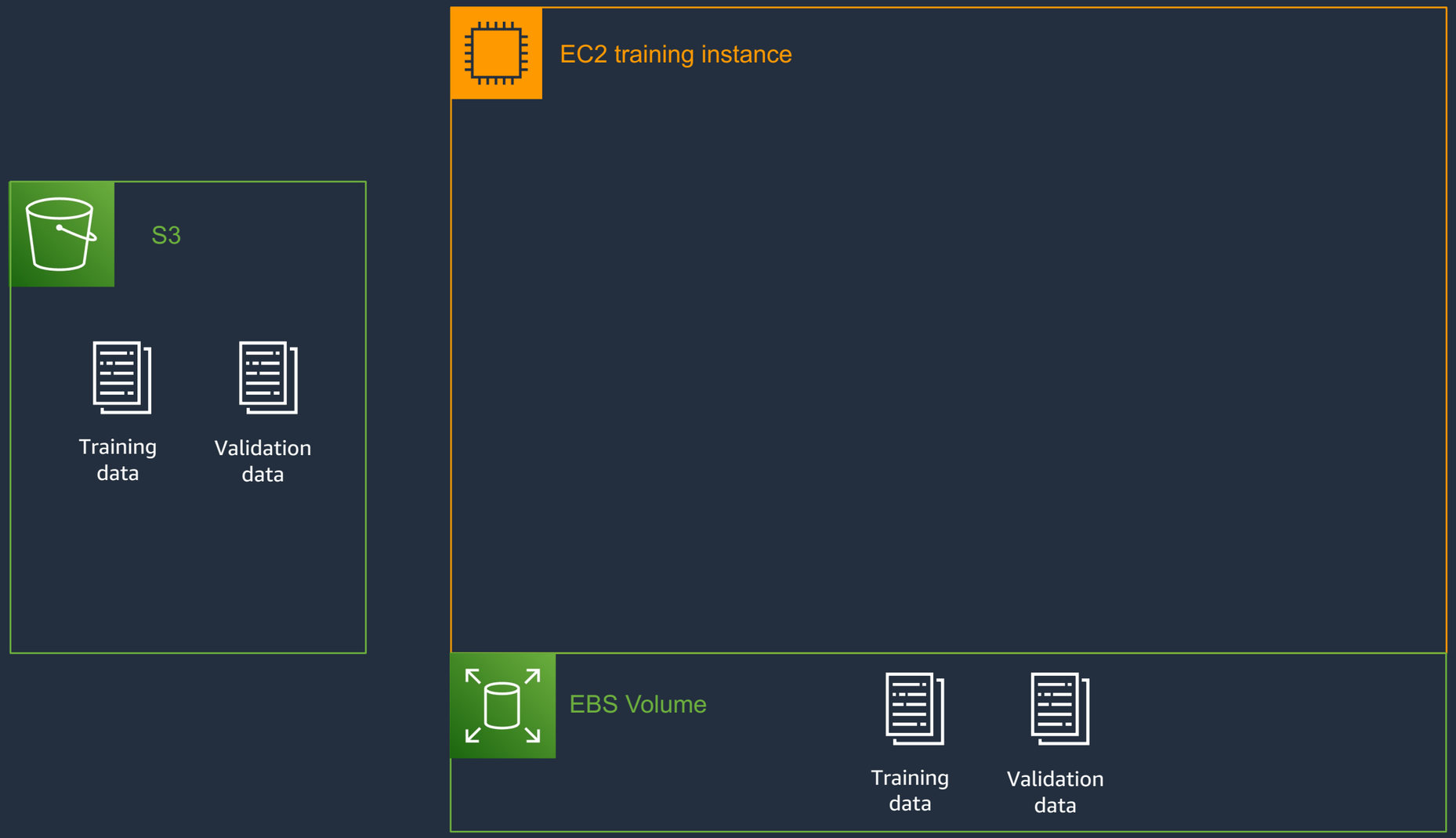
Copy the data from S3 to EBS volume of the EC2 instance



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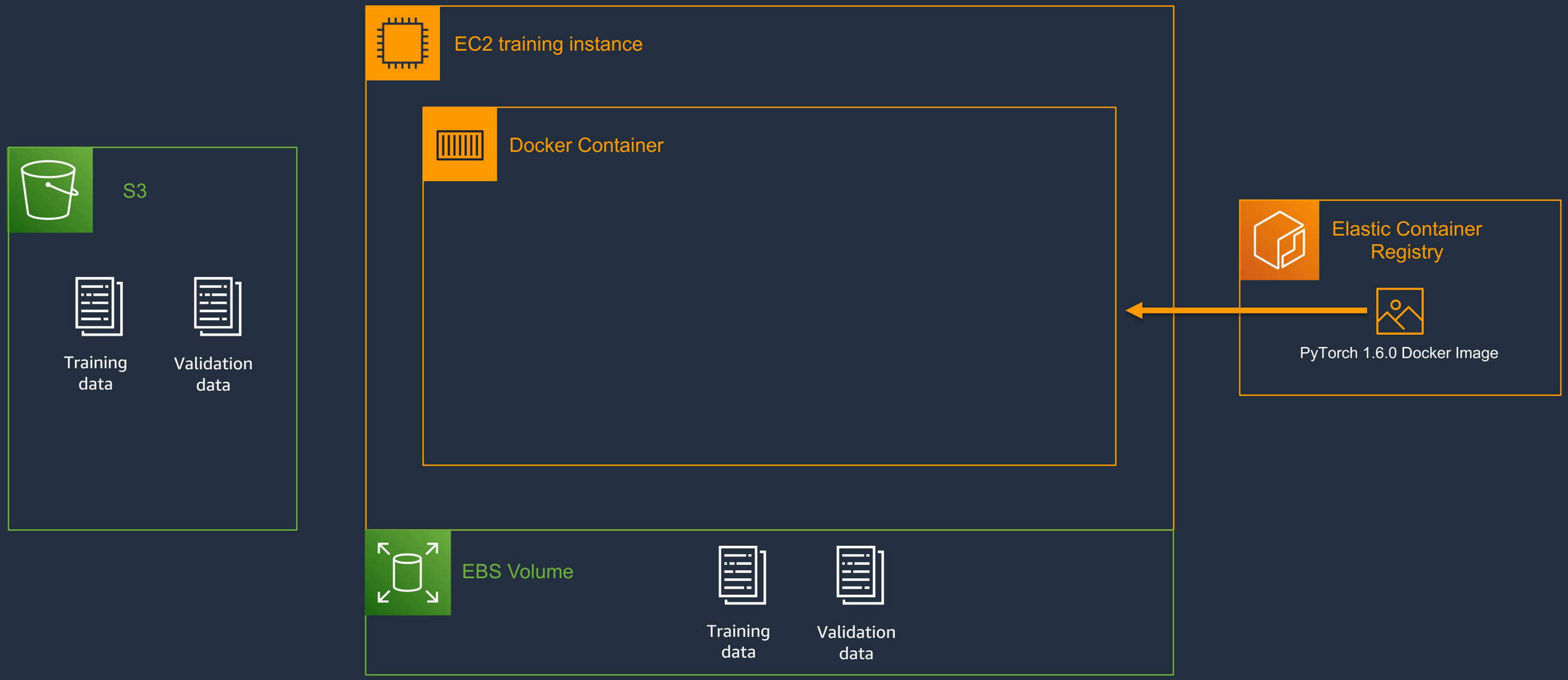
Docker Container



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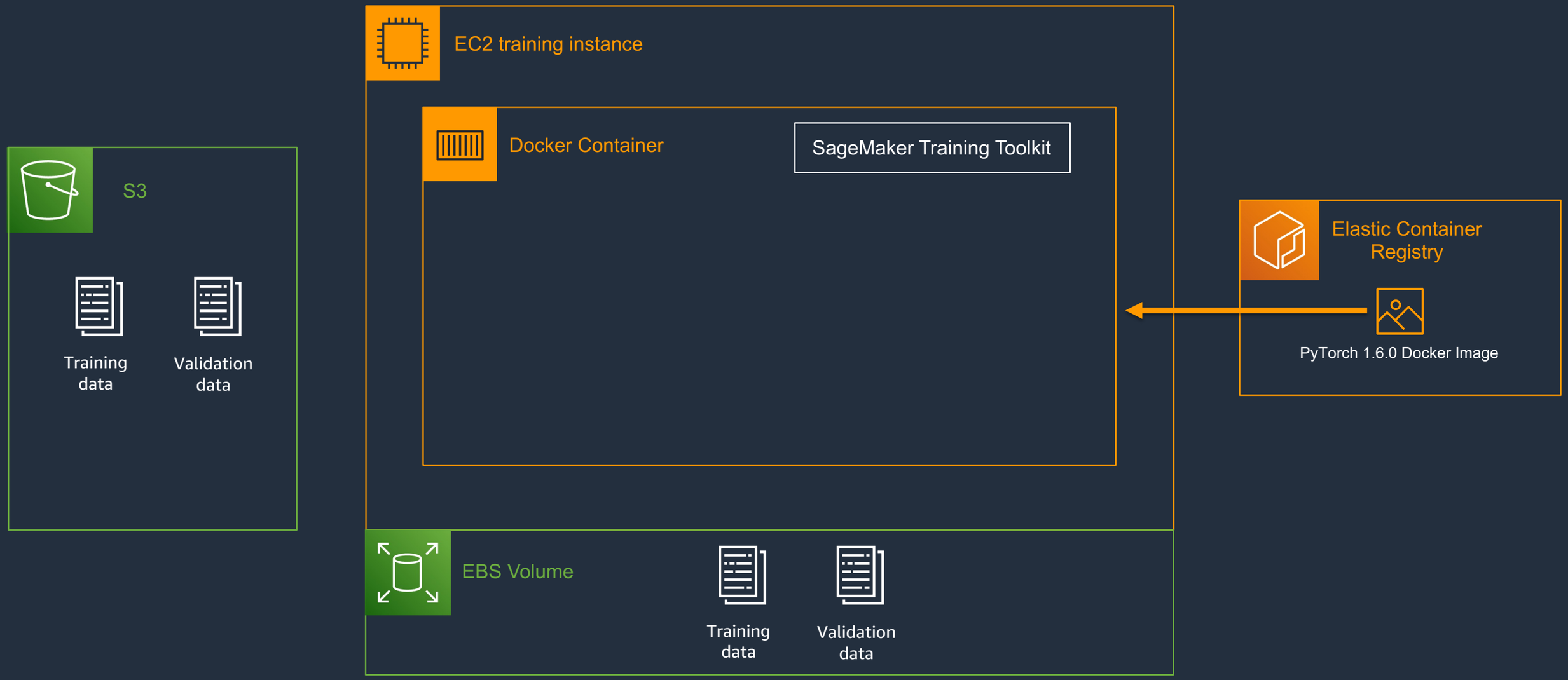
SageMaker Training Toolkit



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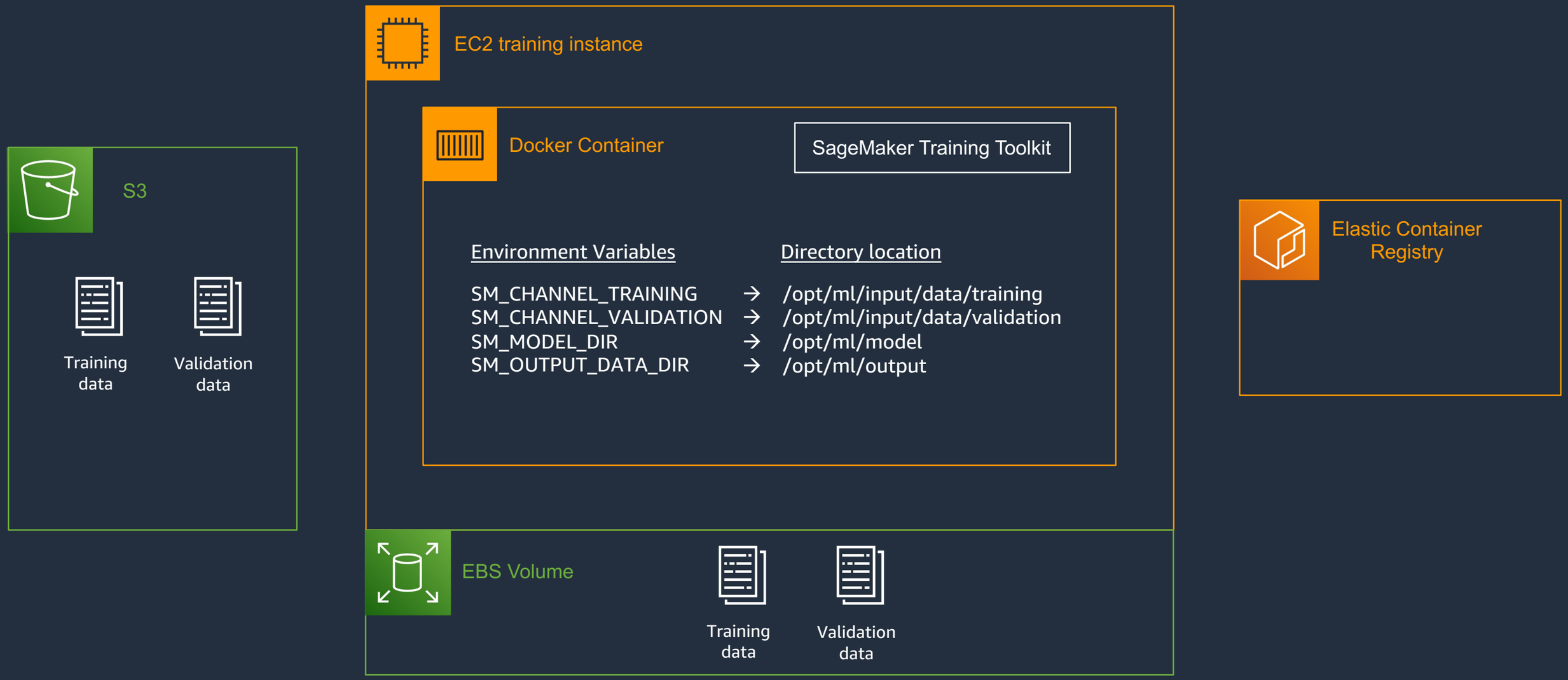
Environment variables and directory location



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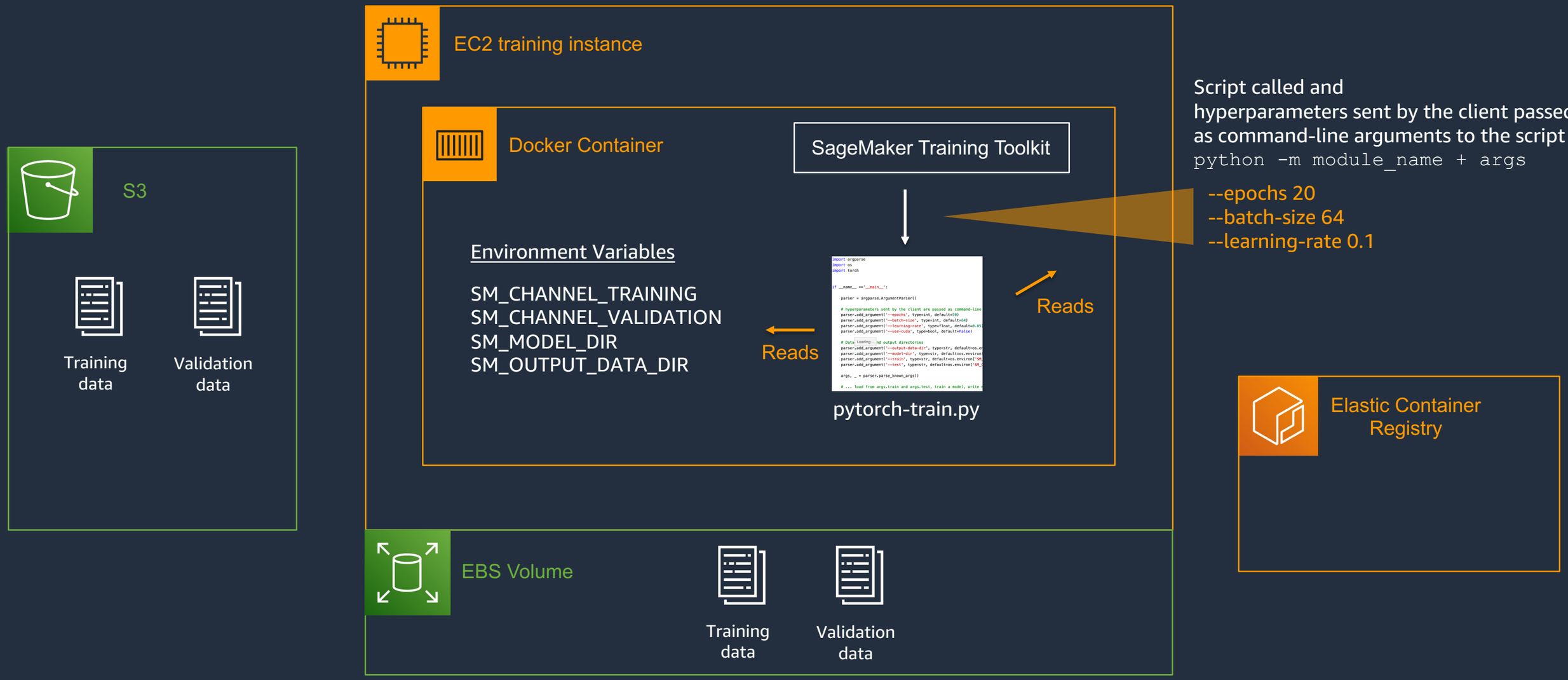
Your script is called with arguments present in the Estimator



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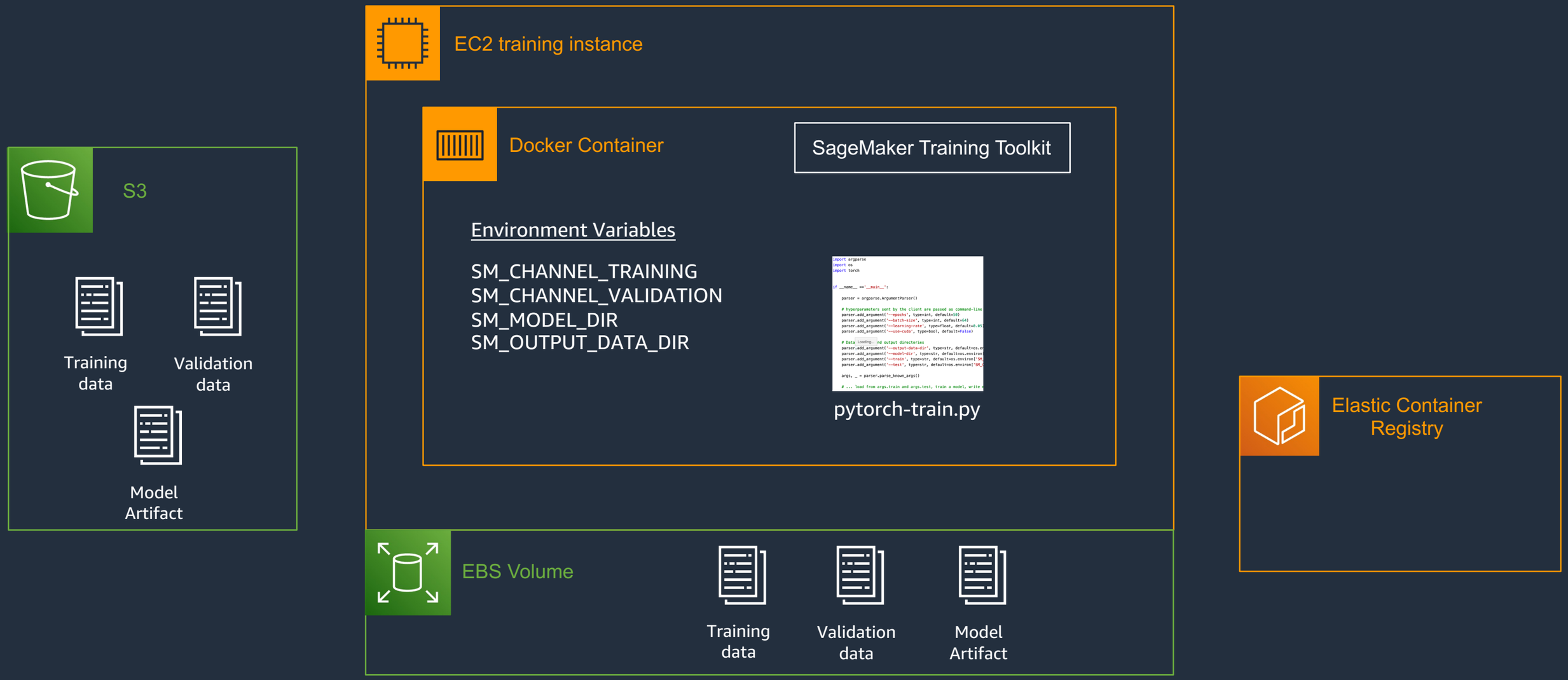
Copy of the Model Artifact to S3 after training



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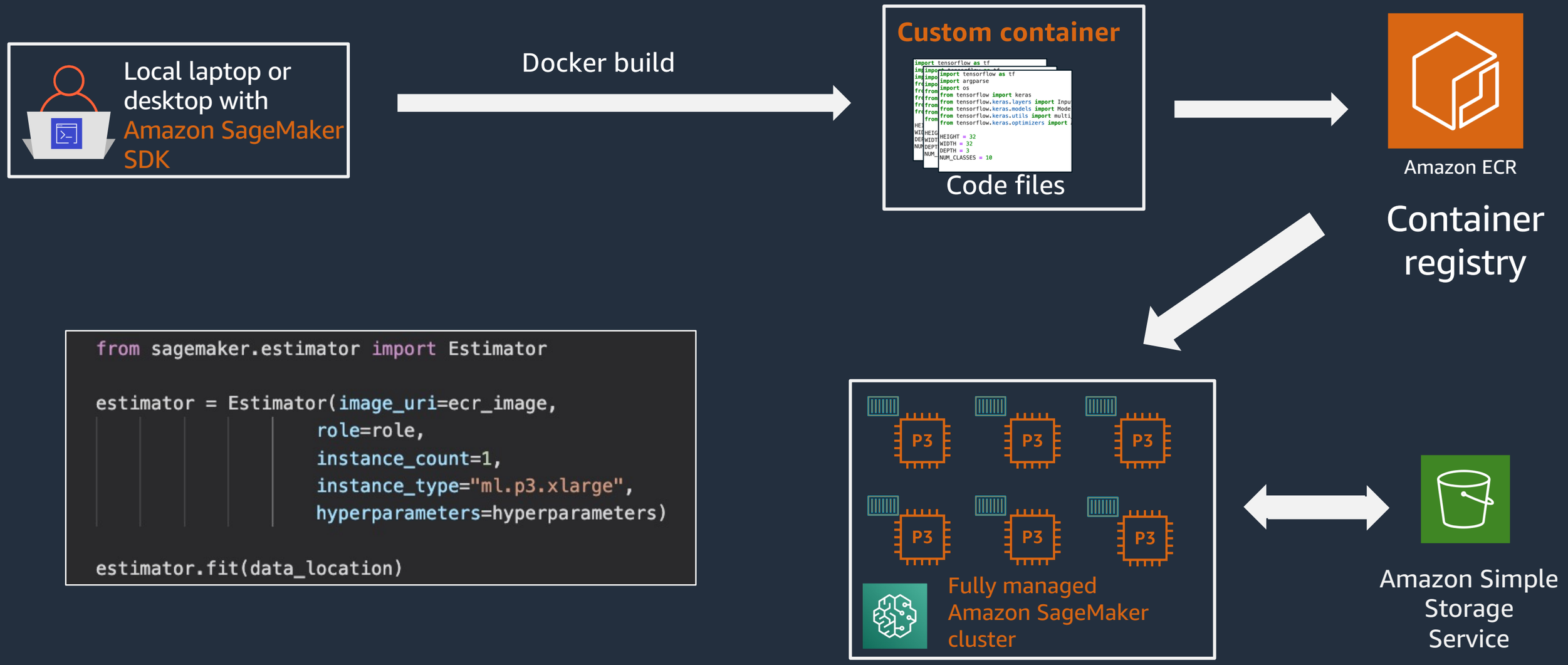


```
from sagemaker.estimator import Estimator

estimator = Estimator(image_uri=ecr_image,
                      role=role,
                      instance_count=1,
                      instance_type="ml.p3.xlarge",
                      hyperparameters=hyperparameters)

estimator.fit(data_location)
```

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Bring Your Own Container

Customer Managed

1. Write your model however you please
2. Point to your model within your Docker file
3. Register your container on ECR
4. Point to your container's address in ECR
5. Don't forget to implement a serve() function!

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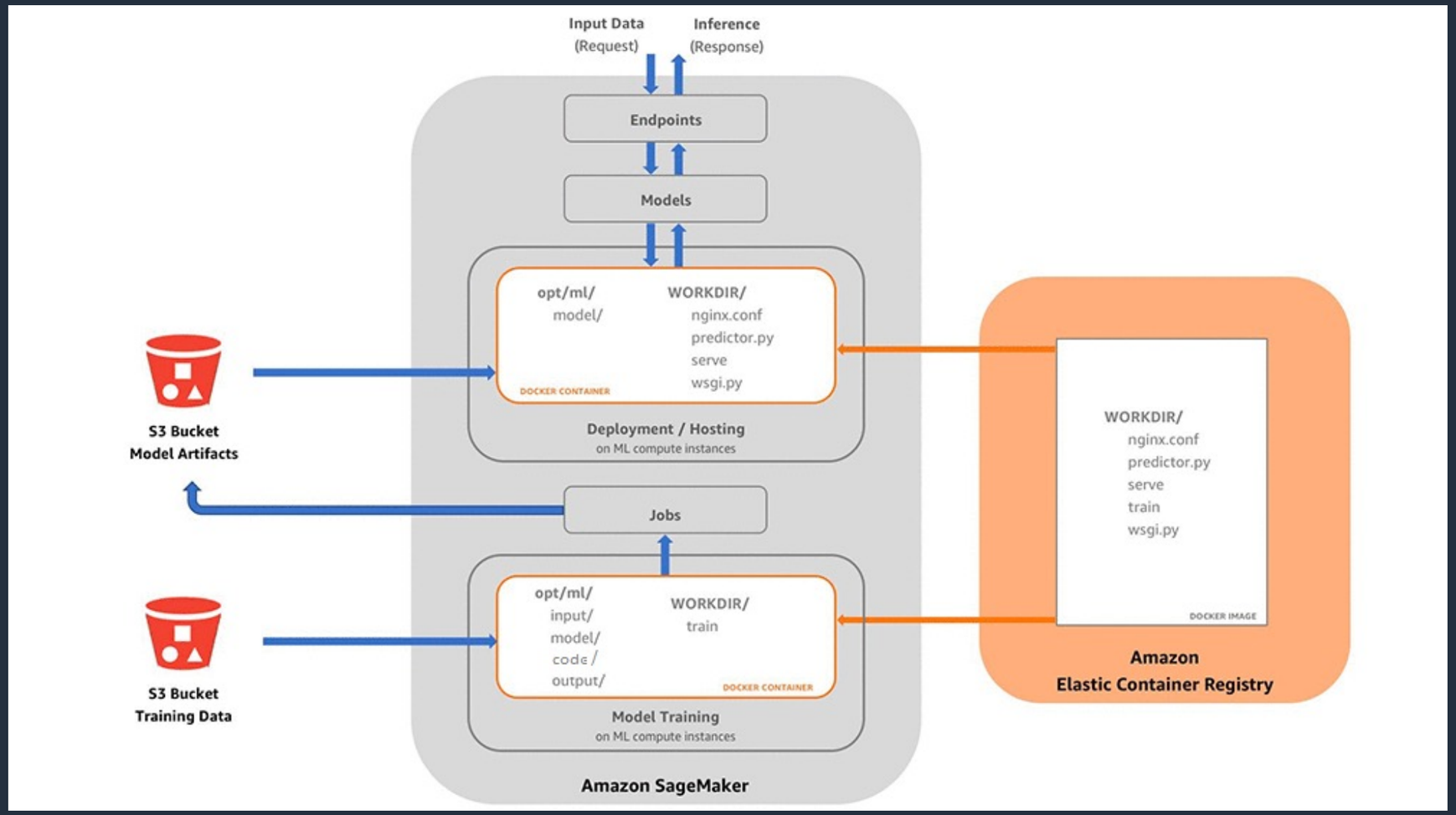
Inside your docker image

Custom Docker image:

- Training image
- Inference (serving) image



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Amazon SageMaker | Training

Bring your own container

2. SageMaker will place hyperparameters here

3. Input (training) data will be connected here

1. Your training code

4. Your code will create model

```
/opt/ml
├── input
│   ├── config
│   │   ├── hyperparameters.json
│   │   └── resourceConfig.json
│   └── data
│       └── <channel_name>
│           └── <input data>
├── model
│
├── code
│   └── <script files>
│
└── output
    └── failure
```

5. Final model from /opt/ml/model will be saved to S3 at the end of training session, before the container is destroyed.

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ML Marketplace

Algorithms

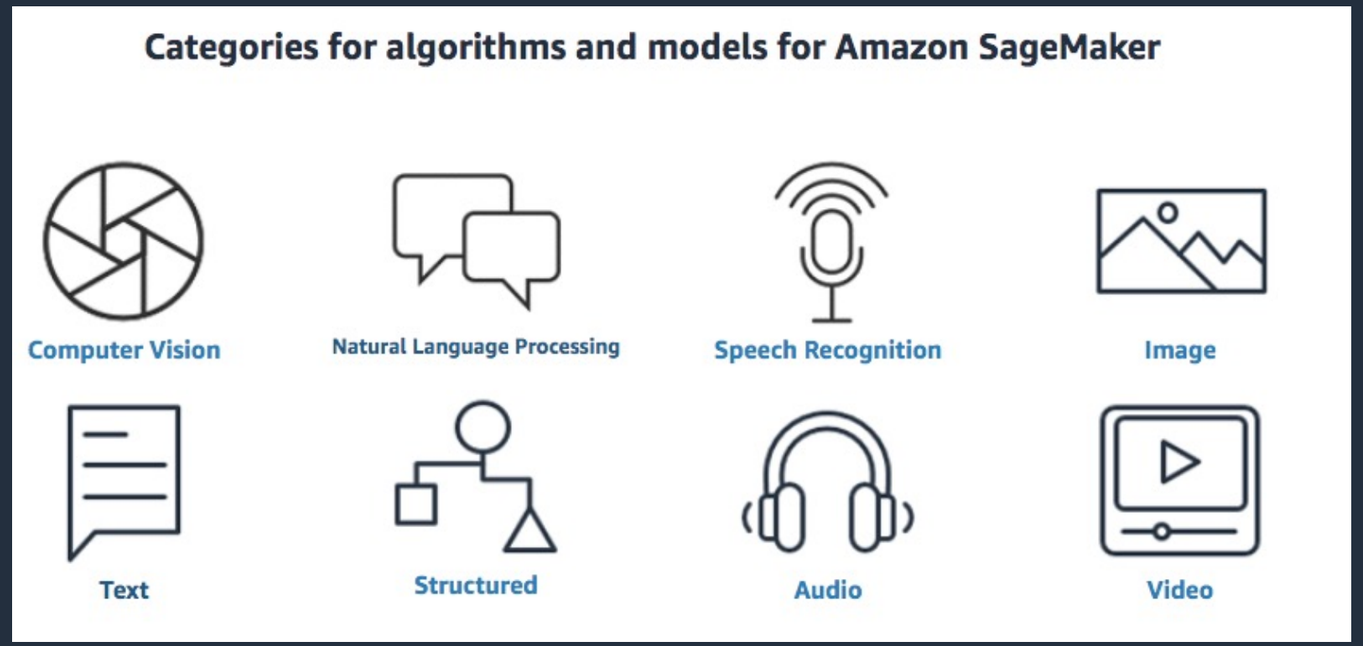
- You can train on your data

Models

- Use a pretrained model artifact

- Subscription model
- Free tier!

230+ solutions!



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Pro Tips

	Script Mode	BYO Docker	ML Marketplace
Pros	Quickly train your own model	Most flexible	Huge variety: easy to add value quickly
Cons	Limited to managed options	More time consuming	Less insight into solution

SageMaker reads from /opt/ml

Find an example to modify

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Work thru the BYO script Notebook Back at 3pm

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